

Note: This exercise is to be done in **groups**. Submission is a .zip file.

Your final exercise is to simulate video streaming using a producer-consumer paradigm.

Since video encoding and decoding are beyond the scope of our class, we shall use plain text files that encode the video frames as ASCII art. A sample movie text file can be downloaded from our Canvas website. The file format is straightforward:

- Each frame begins with a special escape character sequence: the ASCII escape character ESC (value of 27 in decimal or 033 in octal), followed by the character 'c'. On UNIX systems, printing this sequence of characters happens to clear the screen.
- Each frame contains *n* lines, each separated by the `\n` character. The actual size of each frame is variable.
- The frame rate of the ASCII video file must be specified separately by the user, but typically they are encoded at 24, 25, or 30 frames per second (FPS). But depending on the processing power of the display device, the video itself may need to be played back at a lower frame rate.

Create a pair of C++ programs called **producer.cpp** and **consumer.cpp**. You must use System V semaphores and shared memory for this exercise!

The **producer** shall:

- (1 pt) Accept two arguments from the command line:
 - The file name of the ASCII video file.
 - The user-specified FPS of the video file (must be a positive integer)
- (3 pts) Read the ASCII video file on a per-frame basis, and write each frame into a *shared memory* region (and only when it has exclusive access to the memory!). The shared memory region should only contain one frame at a time!
- (1 pt) The next frame should only be read after the frame rate time has passed. For example, if the user specifies 25 fps, each frame should be written into the shared memory region every 40 milliseconds.
- (1 pt) When the end of the video file is reached, the program should simply go back to the start of the video and replay it from the beginning, in an infinite loop.
- (1 pt) When the user presses the Enter key, the producer process should exit.

The **consumer** shall:

- (1 pt) Accept one argument from the command line:
 - An alternative FPS value for playback. This may be lower or higher than the frame rate of the producer.
- (3 pts) Read the shared memory region only at the user-specified FPS rate. Output each frame to standard output, starting with the escape sequence to clear the screen.
 - Note that your computer's terminal may need to process frames at a slower rate than what the user specified. Printing out text in the terminal is slow, so don't go overboard with the FPS values that you test your program with!

Submit a zip file containing the following:

- `producer.cpp`
- `consumer.cpp`
- `shared.cpp` (optional; you may include other files such as header files that are shared between the producer and consumer, but please document how you compile each of your programs)
- Do NOT include any movie files!
- accomplished COA: `Lab9_(surnames in alphabetical order)_COA.pdf`
- name the zip file: `Lab9_(surnames in alphabetical order).zip`

Example submission (the other files are contained within the zip file):

- `Lab9_EspinosaManaogSanchez.zip`
 - `producer.cpp`
 - `consumer.cpp`
 - `shared.cpp`
 - `shared.h`
 - `Lab9_EspinosaManaogSanchez_COA.pdf`

(4 pts) There will be a **defense** for this lab. You are also expected to submit a mandatory **peer evaluation** for the semester, which will be a separate submission in Canvas.

As usual, if you did any research, declare your sources.